

Radian measure



1

a $\frac{\pi}{180}$

b $\frac{180}{\pi}$

2

a 2π radians

b π radians

c $\frac{\pi}{2}$ radians

d $\frac{5\pi}{4}$ radians

e $-\frac{5\pi}{3}$ radians

f $\frac{28\pi}{45}$ radians

g $\frac{29\pi}{180}$ radians

h 2.81 radians

i 5.61 radians

j $-\frac{\pi}{4}$ radians

k $\frac{\pi}{4}$ radians

l $\frac{11\pi}{6}$ radians

m $-\frac{\pi}{3}$ radians

n $\frac{2\pi}{3}$ radians

3

a True

b False; 60° must equal to $\frac{\pi}{3}$ radians, because it is $\frac{1}{3}$ of 180° .

c False; 30° must equal to $\frac{\pi}{6}$ radians, because it is half of 60° .

d True

e True

f False; 225° must be equal to $\frac{5\pi}{4}$ radians, because it is five groups of 45° .

4

a 60°

b 30°

c 45°

d 270°

e 225°

f -120°

g -300°

h 240.6°

5

a i 60°

ii 150°

b i $\frac{\pi}{4}$ radians

ii $\frac{3\pi}{4}$ radians

c i $30'$

ii $90^\circ 30'$

d i $\frac{9\pi}{22}$ radians

ii $\frac{10\pi}{11}$ radians



Unit circle

6

a Quadrant 2

b Quadrant 3

c Quadrant 1

d Quadrant 4

7

a Point B

b Point A

c Point D

d Point C

8

a Point A

b Point B

c Point C

d Point D

9

a $\frac{1}{4}$

b $\frac{1}{3}$

c $\frac{2}{7}$

d $\frac{5}{2}$

10

a $\frac{7\pi}{6}, -\frac{5\pi}{6}$

b $\frac{\pi}{3}, -\frac{5\pi}{3}$

c $\frac{11\pi}{6}, -\frac{\pi}{6}$

d $\frac{2\pi}{3}, -\frac{4\pi}{3}$

11

a $\frac{14\pi}{13}$

b $-\frac{12\pi}{13}$